

YESHWANT G.N

Aspiring ML Engineer | Data Analyst | VLSI Design Engineer | Edge AI Developer

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 Portfolio

SUMMARY

Results-driven engineer Pursuing B.E. in Electrical Engineering (VLSI Design), alongside IIT Madras BS in Data Science and an AI/ML Certification from IIT Guwahati. Experienced in building CNN-based inspection systems achieving 94% accuracy, deploying YOLOv8 models at 28 FPS on Raspberry Pi, and developing customer intelligence platforms using Power BI and predictive analytics. Passionate about applying AI and data-driven solutions to semiconductor, manufacturing, and intelligent systems challenges.

PROFESSIONAL EXPERIENCE

Front-End Development Team Lead

Dark Dragons Caelestis Pvt Ltd

08/2025 – Present
Kannigaipair, Tamil Nadu, India

- Progressed through multiple responsibilities within an early-stage startup, contributing as a Technical Research Associate, Python Debugging Engineer, and ultimately Front-End Development Team Lead.
- Conducted technical research and feasibility analysis to support product strategy and technology adoption decisions.
- Diagnosed and resolved application-level Python issues, improving software reliability and reducing debugging turnaround times.
- Led development of a production-grade web platform using React, JavaScript, HTML5, and CSS3, supporting live business operations.
- Refactored reusable UI components and optimized front-end architecture, reducing browser-specific defects by approximately 30%.
- Collaborated with founders and stakeholders to translate business requirements into scalable features, reducing revision cycles by 50%.

EDUCATION

Bachelor of science in Data Science

IIT MADRAS

Trained in DSA | Probability & Stats | DBMS

05/2025 – Present
Guindy, India

B.E.Electrical Engineering (VLSI DESIGN)

J.N.N.INSTITUTE OF ENGINEERING

learning Digital Electronics | VLSI Design | Semiconductor Manufacturing

09/2024 – Present
Kannigaipair, India

Certification Program in Artificial Intelligence & Machine Learning

IIT GUWAHATI

Trained in Developing ML Models | DL, along with Computer vision

05/2025 – 06/2026
Guwahati, India

SKILLS

Programming

Python | SQL | HTML | CSS | JavaScript

Tools

Vercel | Git & GitHub | Google Colab | VS Code |
Power BI | Power Query M | Excel | PowerPoint

Data Science & Analytics

Pandas, NumPy, Power Query, EDA, Feature Engineering, ROC-AUC

AI & Machine Learning

TensorFlow, PyTorch, Scikit-Learn, CNNs, Transfer Learning, YOLOv8,

Hardware & VLSI

Verilog, Digital Logic Design, VLSI Design, Arduino,

CERTIFICATES

Machine Learning with Python | Mindenious | Feb 2026 – Mar 2026

Hands-on training in feature engineering, model development, and model validation using Scikit-learn. Certificate ID: MIND-INT-25F922

Python for Data Science | Infosys Springboard | Jul 2025

Covered NumPy, Pandas, EDA, data preprocessing, statistical analysis, and ML workflow fundamentals.

LANGUAGES

- Telugu
 - English
- Tamil
 - Hindi

PROJECTS

AI-POWERED SMART CHIP FAULT DETECTION SYSTEM | GitHub [↗](#)

Python · TensorFlow · OpenCV

- Framed semiconductor quality inspection as a CNN-based visual defect classification problem on chip surface images.
- Preprocessed image dataset using OpenCV (enhancement, resizing, augmentation) to improve training data quality.
- Trained a CNN model achieving **94% classification accuracy** across fault categories on held-out test data.
- Evaluated with precision, recall, and F1-score; deployed an automated inspection pipeline, reducing manual review effort by ~40%.

E-COMMERCE CUSTOMER INTELLIGENCE PLATFORM | GitHub [↗](#)

Python · Scikit-learn · Pandas · Power BI

- Framed customer analytics as segmentation and purchase prediction problems to support personalized marketing strategies.
- Integrated transactional, behavioral, and demographic customer data and performed data cleaning and feature engineering.
- Applied clustering algorithms for customer segmentation and machine learning models to predict purchase propensity.
- Evaluated model effectiveness using silhouette score, accuracy, ROC-AUC, and business-oriented performance metrics.
- Developed dashboards and insights that enabled targeted campaigns and improved customer engagement.

EDGE AI SYSTEM FOR REAL-TIME OBJECT DETECTION | GitHub [↗](#)

Python · YOLOv8 · OpenCV · Raspberry Pi

- Framed real-time scene understanding as a low-latency object detection problem for edge-constrained hardware.
- Annotated and prepared video frame datasets; trained and quantized a YOLOv8 model for edge deployment.
- Achieved **28 FPS inference** at **mAP@0.5 of 0.81** on Raspberry Pi with optimized latency under 36ms.
- Eliminated cloud dependency, enabling fully offline real-time detection with reduced compute footprint.